NeurIPS 2026 · Competition Track

Fusion Equilibrium Challenge

Can AI reconstruct the magnetic geometry of a fusion plasma — without measuring the magnetic field?

Two challenges · Two $500 awards · Open dataset · **Phase 1 open now**

THE PROBLEM

Future fusion reactors — SPARC, ARC, ITER — will operate under neutron fluxes that degrade magnetic sensors. Without those sensors, traditional plasma control systems lose their primary inputs.

The challenge: reconstruct the full magnetic geometry of a fusion plasma using only non-magnetic diagnostics — the inputs future reactors will still have.

THE DATASET

Machine	Shots | Operator | Location
DIII-D	7,915 shots · General Atomics · San Diego, USA
MAST	1,206 shots · UKAEA · Culham, UK
Total	9,121 shots · 98 GB · CC BY 4.0 on Hugging Face

Real experimental data — not simulations.

TWO CHALLENGES · TWO AWARDS

→ $500 — Best DIII-D intra-machine reconstruction
→ $500 — Best zero-shot DIII-D → MAST cross-machine generalization
→ Named co-authorship on the lessons-learned paper
→ Invited talk at NeurIPS 2026 Competition Workshop

BUILT WITH

General Atomics	DIII-D National Fusion Facility — largest magnetic fusion research facility in the US
UKAEA	UK Atomic Energy Authority — MAST data via FAIR-MAST open data program
UT Austin	Institute for Fusion Studies — research co-organizer
Hugging Face	Dataset hosting — open access, CC BY 4.0
NeurIPS	The world's most cited ML conference — Competition Track 2026

TIMELINE

→ Phase 1 open now → October 18, 2026

→ Phase 2 (blind test) → October 19–26, 2026
→ Results at NeurIPS 2026 → December 2026

For full details, rules, and the official announcement:

🔗 **Challenge:** fusion-equilibrium-challenge.sophelio.io

🔗 **Announcement:** sophelio.io/fusion-equilibrium-challenge-neurips-2026-competition

🔗 **Dataset:** huggingface.co/datasets/Sophelio/fusion-equilibrium-challenge